

Reference excerpt 02-026

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and v , v and y , respectively (see Fig. 2) and are measured by an operator over a selected cardiac cycle. The detection algorithm takes into account their possible modification during subsequent cycles, by means of the parameter (see Eq. (1)). Once identified the waves a_i , c_i , x_i , v_i and y_i on each cardiac cycle, the intervals tax_i and tv_yi are calculated for each cardiac cycle. The parameters defined in Eq. 1 are used to define the two quantities $ai = ai - xi$ $vi = vi - yi$ (2) The average and the standard deviation for each parameter ai , vi , ai , vi , tax_i and tv_yi are calculated over several cardiac cycle. An illustrative JVP trace is presented in Fig.3 where the waves a , c , x , v and y are visible. The parameters ai , xi , vi and yi for each cardiac cycle i were identified by the algorithm. However, about 10 % of the points are normally not properly detected by the algorithm, while it was possible to identify them manually (see Fig. 4). The parameters ai , vi , tax_i and tv_yi were calculated as in Eq. (2), The mean values and standard deviations of the parameters are presented in Table 1. JVP and ECG time relationship The time interval between the wave a and the events P and R and the wave x and the event T are calculated for each cardiac cycle. These intervals are defined as: $taRi = tai - tRi$ $taPi = tai - tPi$ $txTi = txi - tTi$ (3) 2 Table